



Joint Detection of AI-Generated Images and Post-Processing Alterations

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Computer Vision AIRO Project



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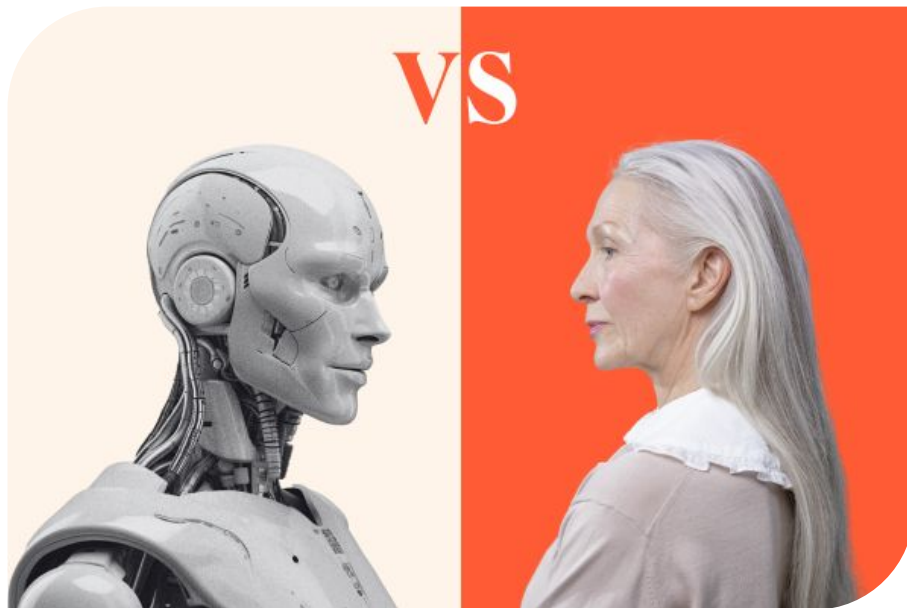
Problem Statement

AI-generated image detection is commonly evaluated under ideal conditions.

However, real-world images shared online are frequently altered by compression, transmission artifacts, and re-digitization processes.

The project addresses two tasks simultaneously:

- **real vs AI-generated** image detection
- **post-processing** transformation classification



State of the Art

Model	Type	Strength	Limitation
ResNet	CNN	Strong baseline	Sensitive to degradation
EfficientNet	CNN	Efficient scaling	Less robust to transformations
ViT	Transformer	Global context	High computational cost
Swin Transformer	Transformer	Hierarchical attention	Requires large datasets
FFT-based	Frequency	Spectral artifact detection	Weak semantic understanding

Model	Original		Transmission		Re-digitization		Overall ACC(%)
	Fake(%)	Real(%)	Fake(%)	Real(%)	Fake(%)	Real(%)	
Detectors (Train on GenImage-SDv1.4 & fine-tune on RRDataset)							
DRCT-ConvB[8]	93.52	95.52	92.82(-0.70)	95.09(-0.43)	64.34(-29.18)	96.22(+0.70)	89.59
DIRE[54]	89.72	98.25	90.34(+0.62)	97.87(-0.38)	1.42(-88.30)	98.89(+0.64)	79.42
DNF[60]	90.62	99.05	90.98(+0.36)	94.45(-4.60)	0.05(-90.57)	99.98(+0.93)	79.19
AIDE[56]	78.95	78.94	74.72(-4.23)	78.75(-0.19)	76.04(-2.91)	83.13(+4.19)	78.42
GramNet[33]	81.34	74.65	79.49(-1.85)	75.69(+1.04)	79.45(-1.89)	62.02(-12.63)	75.44
CNNSpot[53]	72.42	89.09	65.72(-6.70)	88.78(-0.31)	43.12(-29.30)	86.72(-2.37)	74.31
LNP[32]	83.14	89.26	38.23(-44.91)	89.30(+0.04)	31.91(-51.23)	91.05(+1.79)	70.48
Fredect[15]	79.60	75.93	58.13(-21.47)	82.11(+6.18)	46.34(-33.26)	69.95(-5.98)	68.68
NPR[51]	49.21	96.18	28.08(-21.13)	97.13(+0.95)	38.65(-10.56)	92.42(-3.76)	66.95
Fusing[20]	87.24	92.46	7.38(-79.86)	99.04(+6.58)	30.79(-56.45)	73.97(-18.49)	65.15
SAFE[29]	98.29	88.22	0.88(-97.41)	98.85(+10.63)	2.29(-96.00)	98.82(+10.60)	64.56
Freq-Net[50]	76.08	82.18	4.47(-71.61)	98.64(+16.46)	37.48(-38.60)	77.99(-4.19)	62.81
F3Net[55]	65.82	71.35	52.18(-13.64)	75.49(+4.14)	31.16(-34.66)	74.85(+3.50)	61.81
UnivFD[38]	64.79	64.90	44.61(-20.18)	70.80(+5.90)	36.15(-28.64)	75.69(+10.79)	59.49
C2P-CLIP[49]	17.21	97.54	28.82(+11.61)	99.58(+2.04)	18.01(+0.80)	90.29(-7.25)	58.58
SSP[9]	61.29	64.54	40.62(-20.67)	70.33(+5.79)	32.58(-28.71)	79.94(+15.40)	58.22
LGrad[48]	51.00	81.29	18.86(-32.14)	92.54(+11.25)	14.71(-36.29)	88.27(+6.98)	57.78
VLMs (Zero-shot)							
GPT-4o-latest[1]	96.30	92.68	79.41(-16.89)	90.01(-2.67)	69.23(-27.07)	76.92(-15.76)	84.09
Claude-3.7-sonnet[2]	85.12	94.57	71.26(-13.86)	96.17(+1.60)	62.34(-22.78)	85.41(-9.16)	82.48
Gemini-2-flash[52]	72.10	98.43	52.19(-19.91)	97.41(-1.02)	46.11(-25.99)	97.41(-1.02)	77.28
Grok-2-vision[17]	46.15	91.84	52.12(+5.97)	94.03(+2.19)	48.01(+1.86)	81.63(-10.21)	68.96
Gemini-1.5-pro[52]	36.12	97.78	36.36(+0.24)	95.83(-1.95)	22.22(-13.90)	88.64(-9.14)	62.83
Qwen2vl-72B[3]	31.14	88.74	22.87(-8.27)	89.95(+1.21)	26.99(-4.15)	92.55(+3.81)	58.71
GLM4v-plus[16]	22.16	90.57	30.21(+8.05)	86.01(-4.56)	38.16(+16.00)	82.14(-8.43)	58.21
Moonshot-vision[37]	14.68	99.72	16.24(+1.56)	94.75(-4.97)	24.37(+9.69)	96.84(-2.88)	57.77
Hunyuan-vision[19]	28.57	91.84	20.13(-8.44)	86.06(-5.78)	32.14(+3.57)	81.63(-10.21)	56.73
YI-vision[58]	22.73	89.13	20.45(-2.28)	81.81(-7.32)	28.89(+6.16)	78.26(-10.87)	53.55

Proposed Method

The proposed framework uses a shared **CNN** backbone with two task-specific classification heads.

A pretrained **ResNet18** extracts shared visual features, which are simultaneously used for:

- binary real/fake classification
- transformation classification

Joint optimization is performed using a weighted multi-task loss. Different weights were evaluated to analyze the trade-off between tasks.

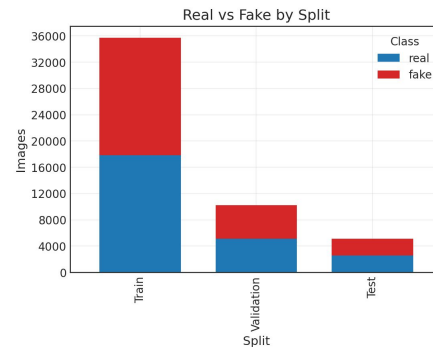
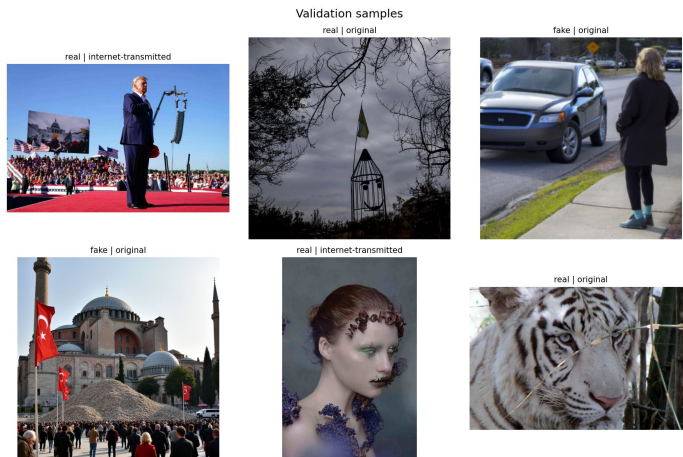
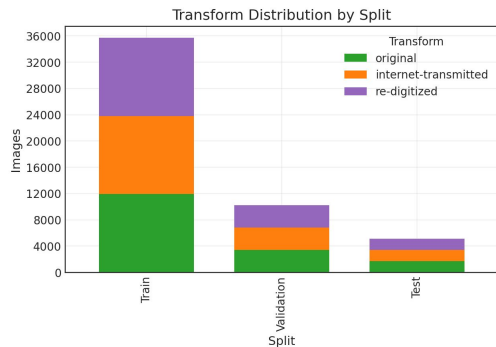
- Weighted **cross-entropy** optimization

Formula	Meaning	Short Explanation
$\hat{y}_{rf} = f_{rf}(x)$	Real/Fake Prediction	Predicts whether the image is real or AI-generated.
$\hat{y}_{tr} = f_{tr}(x)$	Transformation Prediction	Predicts the transformation category.
$f = \text{Backbone}(x)$	Feature Extraction	Shared features extracted by ResNet18.
$z_{rf} = W_{rf}f + b_{rf}$	Real/Fake Head	Maps features to binary logits.
$z_{tr} = W_{tr}f + b_{tr}$	Transformation Head	Maps features to transformation logits.
$\mathcal{L}_{rf} = -\sum_i y_i^{rf} \log(\hat{y}_i^{rf})$	Real/Fake Loss	Binary classification loss.
$\mathcal{L}_{tr} = -\sum_i y_i^{tr} \log(\hat{y}_i^{tr})$	Transformation Loss	Multi-class classification loss.
$\mathcal{L} = \lambda_{rf}\mathcal{L}_{rf} + \lambda_{tr}\mathcal{L}_{tr}$	Multi-Task Loss	Weighted sum of both losses.
$\text{score} = 0.5 (\text{Acc}_{rf} + \text{Acc}_{tr})$	Validation Score	Combined metric for model selection.
$F(x) + x$	Residual Connection	Skip connection in ResNet18.
$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$	Softmax	Converts logits into probabilities.

Dataset

The project uses **RRDataset**, a benchmark designed for robust AI-generated image detection under realistic conditions.

The dataset contains: **real** photographs | **AI-generated** images | multiple **transformation** categories (*internet, re-digitized, original*).



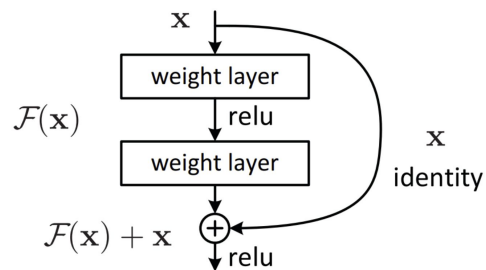
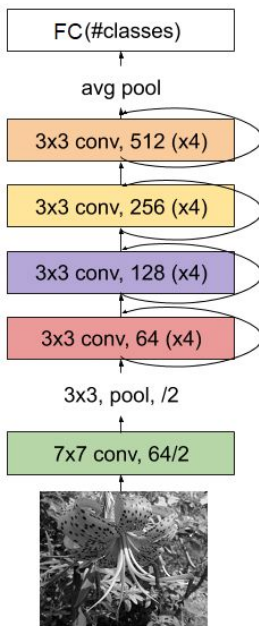
Experimental Setup

Training settings

- Optimizer: *AdamW*
- Image size: 224×224
- Batch size: 16
- Epochs: 5
- Learning rate: $1e-4$
- Weight decay: $1e-4$
- Dropout: 0.3

Evaluation metrics

- Accuracy
- F1-score
- Joint correctness



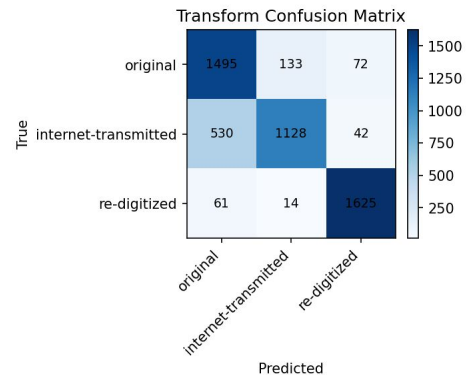
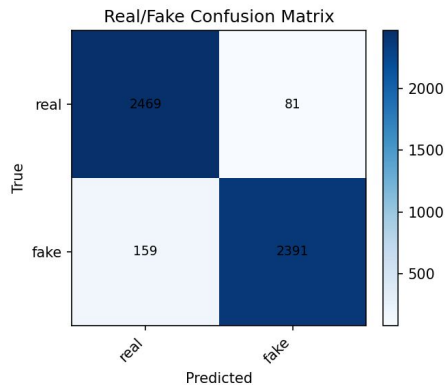
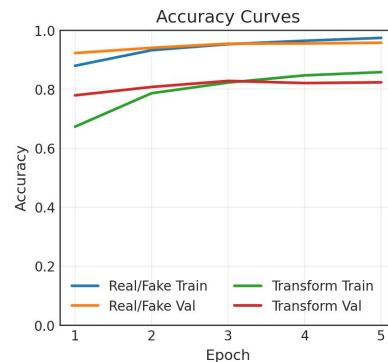
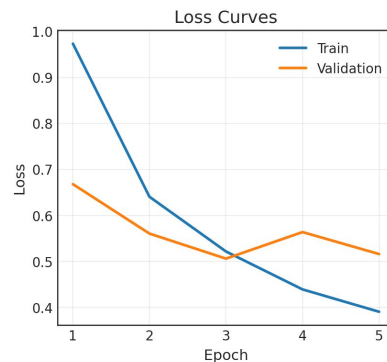
Model Evaluation

Final performance

- Real/Fake Accuracy $\approx 95.3\%$
- Transformation Accuracy $\approx 83.3\%$
- Real/Fake Macro F1 ≈ 0.952
- Transformation Macro F1 ≈ 0.831

Key findings

- Binary detection remains highly robust
- Transformation-related degradations significantly affect detection performance.



Conclusions

Although the model obtained promising results, transformation classification remains more challenging than binary real/fake detection, particularly under strong degradations.

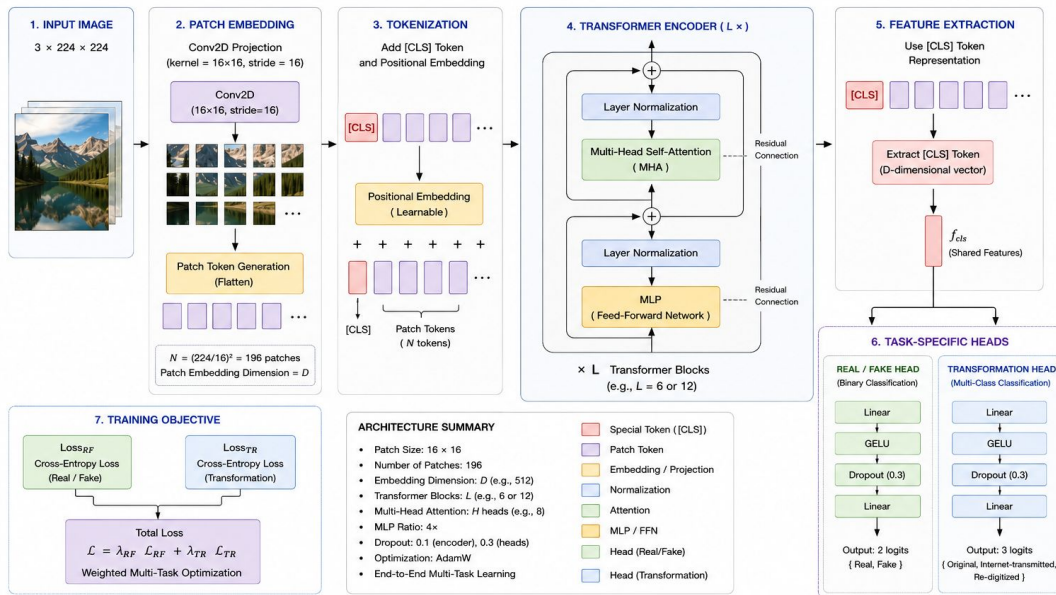
Future work may explore **transformer**-based architectures and larger-scale training to improve robustness.

We were already working on a **ViT** version of the architecture.

**image generated by AI based on our hand drawn scheme + python ViT file*

Multi-Task Vision Transformer (ViT) Architecture

Joint AI-Generated Image Detection and Post-Processing Transformation Classification



This diagram illustrates the complete architecture of the proposed ViT model for joint AI-generated image detection and post-processing transformation classification. MHA: Multi-Head Attention | FFN: Feed-Forward Network | L: Number of Transformer Blocks

References

Papers

- Li, Chunxiao et al. *Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection in Challenging Scenarios*. ICCV 2025.
- Shao, Rui et al. *Detecting and Grounding Multi-modal Media Manipulation*. CVPR 2023.

Tools & Frameworks

- PyTorch & torchvision
- scikit-learn
- matplotlib
- NumPy